# Investigation: MC Purity Correction Factor in PPG12

PPG12 Analysis Team (automated investigation)

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## 1 Introduction

The PPG12 isolated photon cross-section measurement employs the ABCD method to estimate photon purity from data. The four ABCD regions are defined by two axes: a BDT-based tight/non-tight classification and an isolation/non-isolation criterion. Signal (isolated photons) populates region A (tight, isolated), and the background contribution to A is estimated from the other three regions as $N^A_{\text{bkg}} = R \cdot (B \times C)/D$, after correcting for signal leakage into regions B, C, and D (fractions $c_B$, $c_C$, $c_D$).

In Monte Carlo, the truth purity can be computed directly by matching reconstructed clusters to generator-level photons. The ratio of truth purity to the ABCD-estimated purity (with leakage corrections applied) provides a multiplicative correction factor. This factor is stored as `g_mc_purity_fit_ratio` and applied to the data purity when the configuration flag `mc_purity_correction: 1` is set.

A notable feature of this correction factor is that it crosses 1.0 in the neighborhood of $p_T \approx$ 13–14 GeV. Below this crossing point, the ABCD method underestimates purity (correction $> 1$); above it, the ABCD method overestimates purity (correction $< 1$). This report documents the investigation into the origin of this behavior.

**Important technical detail:** the stored ratio uses *truth purity data points* divided by a *smooth Padé fit* to the ABCD purity (`f_purity_leak_fit`), not a ratio of two fitted curves. Consequently, the correction factor inherits the statistical fluctuations of the truth purity measurement. When applied to data via `TGraphErrors::Eval()`, the correction uses linear interpolation between these noisy stored points.

## 2 The Correction Factor

Table 1 lists the MC truth purity, the ABCD purity (with leakage corrections), and the stored correction ratio at each $p_T$ bin center.

Table 1: MC purity correction ratio at each $p_T$ bin. The ratio is truth purity (data points) divided by the Padé fit to the leakage-corrected ABCD purity.

| $p_T$ (GeV) | Truth Purity | ABCD Purity (leak) | Ratio |
|---|---|---|---|
| 9 | 0.566 | 0.370 | 1.505 |
| 11 | 0.576 | 0.460 | 1.206 |
| 13 | 0.620 | 0.654 | 1.112 |
| 15 | 0.594 | 0.640 | 0.952 |
| 17 | 0.584 | 0.630 | 0.862 |
| 19 | 0.622 | 0.620 | 0.860 |
| 21 | 0.667 | 0.700 | 0.875 |
| 23 | 0.679 | 0.804 | 0.852 |
| 25 | 0.651 | 0.560 | 0.787 |
| 27 | 0.707 | 0.915 | 0.829 |
| 30 | 0.679 | 0.712 | 0.766 |
| 34 | 0.849 | 0.889 | 0.919 |

The ABCD purity points are highly non-monotonic (e.g., 0.560 at $p_T = 25$ GeV then 0.915 at $p_T = 27$ GeV), reflecting both the ABCD method's limitations and poor MC statistics at high $p_T$. The truth purity is comparatively stable ($\sim$0.57–0.68) across 9–30 GeV, rising to 0.85 only at 34 GeV.

Figure 1 shows the fit curves, residuals, and ratio.

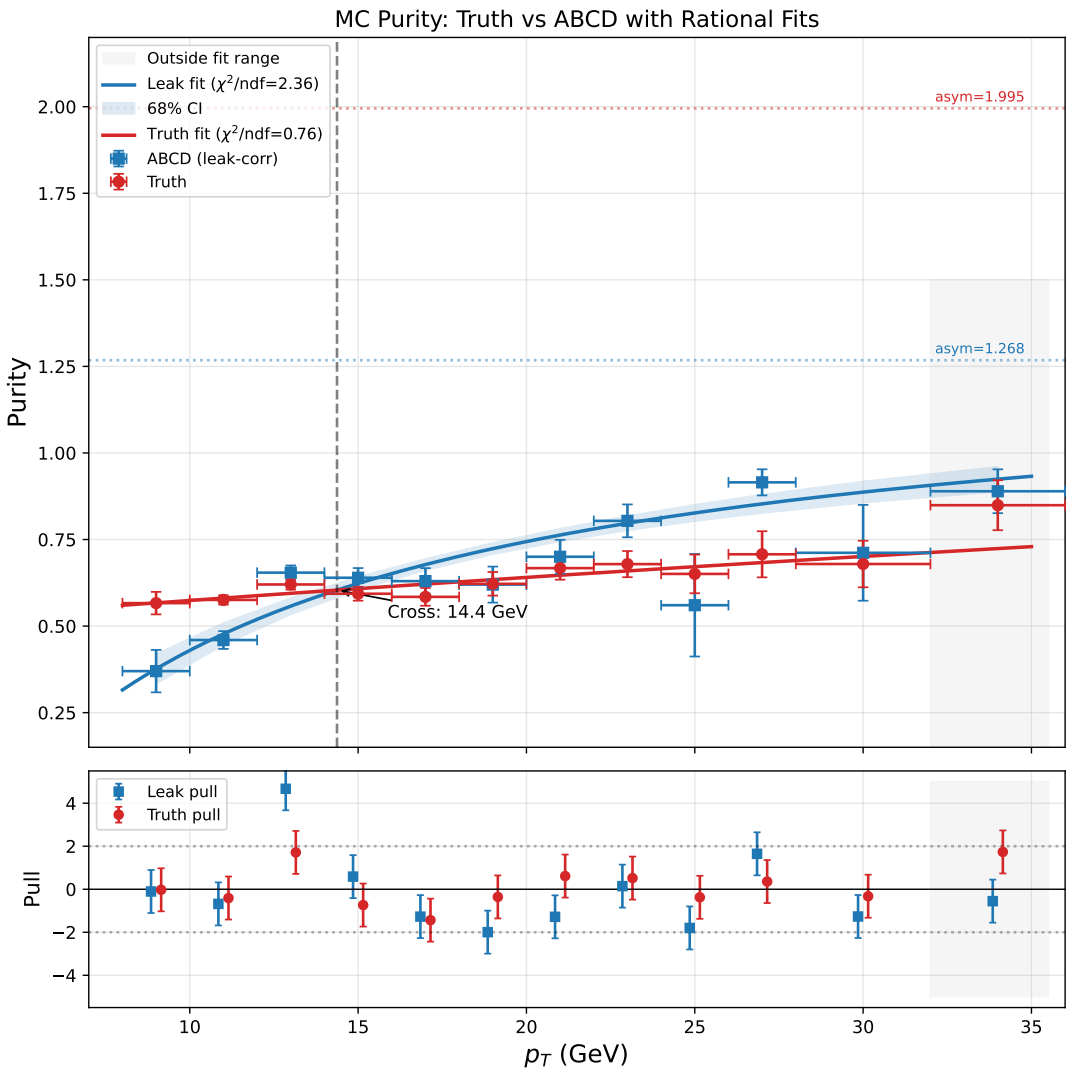


Figure 1: Padé fits to truth and ABCD purity (page 1 of the diagnostic PDF).

# 3 Root Causes

## 3.1 ABCD Independence Violation (Primary Cause)

The ABCD method assumes that BDT tight/non-tight classification and isolation are uncorrelated for background. This assumption is violated.

The true background $R$-factor, defined as

$$R_{\text{true}} = \frac{A_{\text{bkg}} \cdot D_{\text{bkg}}}{B_{\text{bkg}} \cdot C_{\text{bkg}}}, \tag{1}$$

should equal 1.0 if the two axes are independent for background. In practice, $R_{\text{true}}$ varies strongly with $p_T$ (Table 2).

Table 2: True background $R$-factor from MC (from `h_R` in the output file). Values computed from jet MC background after truth-matching subtraction. Fluctuations at high $p_T$ (e.g., $R = 0.76$ at $[24, 26]$ GeV followed by $R = 2.09$ at $[26, 28]$ GeV) reflect limited MC statistics in those bins.

| $p_T$ bin (GeV) | $R_{\text{true}}$ | Deviation from 1.0 |
|---|---|---|
| $[8, 10]$ | 0.70 | $-30\%$ |
| $[10, 12]$ | 0.80 | $-20\%$ |
| $[12, 14]$ | 1.07 | $+7\%$ |
| $[14, 16]$ | 1.14 | $+14\%$ |
| $[16, 18]$ | 1.16 | $+16\%$ |
| $[18, 20]$ | 0.98 | $-2\%$ |
| $[20, 22]$ | 1.04 | $+4\%$ |
| $[22, 24]$ | 1.59 | $+59\%$ |
| $[24, 26]$ | 0.76 | $-24\%$ |
| $[26, 28]$ | 2.09 | $+109\%$ |
| $[28, 32]$ | 0.93 | $-7\%$ |
| $[32, 36]$ | 1.07 | $+7\%$ |

$R_{\text{true}}$ crosses 1.0 in the $[12, 14]$ GeV bin—precisely where the purity correction ratio crosses 1.0. The mechanism is:

- Below $p_T = 13$ GeV: $R_{\text{true}} < 1 \Rightarrow$ ABCD overestimates background $\Rightarrow$ purity too low $\Rightarrow$ correction $> 1$.
- Above $p_T = 13$ GeV: $R_{\text{true}} > 1 \Rightarrow$ ABCD underestimates background $\Rightarrow$ purity too high $\Rightarrow$ correction $< 1$.

An independent test confirms the violation: the isolation fraction for tight-BDT background divided by that for non-tight-BDT background ranges from 0.77 (at low $p_T$) to 1.77 (at $p_T \sim 23$ GeV), where it should be 1.0 if BDT and isolation were independent.

### 3.2 ET-Dependent Tight BDT Threshold

The tight BDT threshold is $E_T$-dependent: $\text{BDT}_{\text{min}}^{\text{tight}} = 0.80 - 0.015 \times E_T$, while the non-tight BDT window is fixed at [0.02, 0.50]. This creates a "dead zone" of BDT scores between the two regions that shrinks with $E_T$:

Table 3: BDT phase-space partition vs. $E_T$.

| $E_T$ (GeV) | Tight threshold | Dead zone | Width |
|---|---|---|---|
| 8 | 0.68 | [0.50, 0.68] | 0.18 |
| 12 | 0.62 | [0.50, 0.62] | 0.12 |
| 15 | 0.575 | [0.50, 0.575] | 0.075 |
| 20 | 0.50 | vanishes | 0 |
| 25 | 0.425 | overlap | — |

At $E_T = 20$ GeV the dead zone vanishes; above that, the tight threshold drops below the non-tight upper boundary, effectively shrinking the non-tight window. This means the background composition in each ABCD region changes with $p_T$, creating the observed $p_T$-dependent BDT–isolation correlation.

This is **not** caused by the BDT model boundary at $E_T = 15$ GeV: both $E_T$ bins use the same model (`base_v1E`).

### 3.3 Fit Quality Issues (Amplifying Factor)

The ABCD purity is fitted with a Padé function $f(x) = (p_0 + p_1 x)/(1 + p_2 x)$ using 11 of the 12 available $p_T$ bins (the last bin at $p_T = 34$ GeV lies outside the fit range [8, 32] GeV):

- Leak fit: $\chi^2/\text{NDF} = 18.9/8 = 2.36$, with a pull of 4.67 at $p_T = 13$ GeV (poor fit quality).
- Truth fit: $\chi^2/\text{NDF} = 6.1/8 = 0.76$ (good fit quality).
- The Padé form forces a monotonic rise, but the ABCD points are non-monotonic (e.g., 0.56 at $p_T = 25$, 0.92 at $p_T = 27$ GeV).

Because the stored correction uses truth *points* divided by the smooth leak *fit*, the fit quality primarily affects the denominator. The poor leak fit means the denominator may be systematically too high or too low at individual $p_T$ bins.

### 3.4 Statistical Limitations at High $p_T$

The effective number of events in region D (the denominator of $B \times C/D$) drops rapidly at high $p_T$:

Table 4: Effective statistics in ABCD region D from jet MC.

| $p_T$ (GeV) | $D_{\text{eff}}$ |
|---|---|
| $< 14$ | $> 1000$ |
| 23 | 107 |
| 25 | 71 |
| 27 | 19 |
| 34 | 7 |

Above $p_T = 25$ GeV, the ABCD estimate is statistically unreliable. This explains the wild fluctuations in the ABCD purity at high $p_T$ and correspondingly large uncertainties on the correction factor.

## 4 Leakage Fractions

The signal leakage fractions (computed from signal MC) are small and well-behaved:

- $c_B = 5$–$13\%$ (signal leaking to tight + non-isolated)
- $c_C = 2.6$–$4.6\%$ (signal leaking to non-tight + isolated)
- $c_D < 0.6\%$ (signal leaking to non-tight + non-isolated)

The leakage correction moves the ABCD purity in the right direction but cannot compensate for the fundamental independence violation described in Section 3.1.

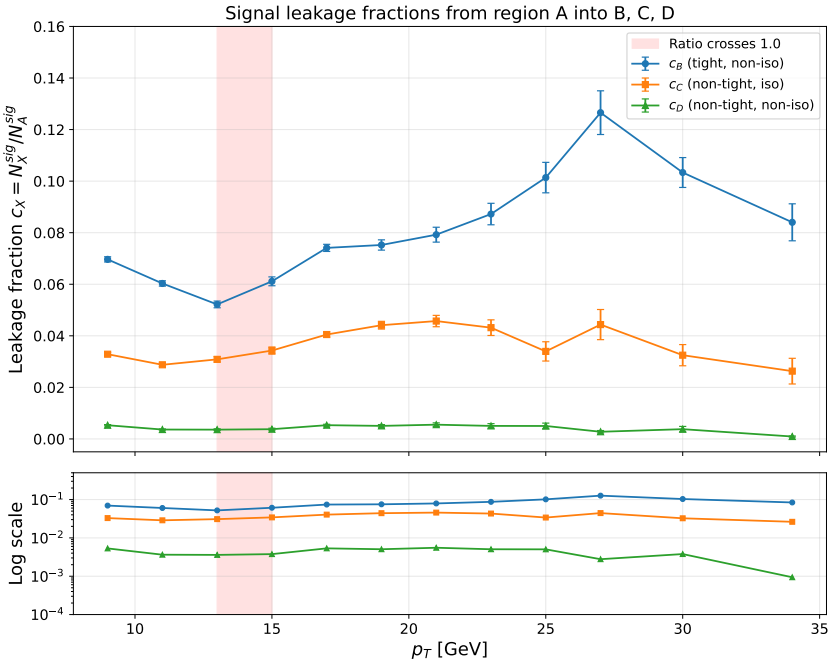


Figure 2: Signal leakage fractions $c_B$, $c_C$, $c_D$ vs. $p_T$.

## 5 Methodology Note

This investigation was conducted using an automated multi-wave agent pipeline:

1. **Wave 1** (3 parallel agents): fit diagnostics, leakage fractions, ABCD components analysis.
2. **Wave 2** (3 parallel agents): independent review of each Wave 1 agent.
3. **Wave 3**: corrections applied based on review findings.
4. **Wave 4–5**: report writing and review.

The Wave 2 review identified and corrected one significant error: the leakage investigation agent initially computed signal fractions from the photon MC file (`MC_efficiency_bdt_nom.root`, which is pure signal by construction), rather than from the jet MC file (`MC_efficiency_jet_bdt_nom.root`, which serves as "data" in MC closure). The signal fractions reported in that agent's analysis (65–77% in region B) were therefore artifactual. The corrected analysis uses `h_R` from the final output file, which is computed from the jet MC with proper background subtraction.

## 6 Implications and Recommendations

1. The MC purity correction addresses a **real bias** (ABCD independence violation), but the correction factor has large statistical uncertainties from truth purity, especially at high $p_T$.
2. The sign change at $p_T \sim 14$ GeV is a **genuine physics effect** ($R_{\text{true}}$ crossing 1.0), not purely a fit artifact.
3. **Consider**: using `h_R` from MC to correct the ABCD $R$-factor directly, rather than correcting the final purity post-hoc.
4. The high-$p_T$ bins ($> 25$ GeV) have very poor MC statistics—the correction factor there is unreliable ($D_{\text{eff}} < 71$).

5. The $E_T$-dependent tight BDT threshold is a design choice that introduces $p_T$-dependent correlations; a fixed threshold would reduce this effect but may sacrifice signal efficiency.

# 7 Re-evaluation: ET-Dependent Non-Tight BDT Bound

*Added April 8, 2026.* The non-tight BDT upper bound was changed from a fixed value of 0.50 to an $E_T$-dependent parametric form matching the tight lower bound: $0.80 - 0.015 \times E_T$. This eliminates the dead zone (Section 3.2) at all $E_T$. The BDT axis is now a clean two-region partition: non-tight = [0.02, threshold), tight = [threshold, 1.0].

## 7.1 Updated Correction Factor

Table 5: MC purity correction ratio: old (fixed non-tight = 0.50) vs. new ($E_T$-dependent non-tight). Truth purity is from generator matching; ABCD purity from leakage-corrected data points (`gpurity_leak`). The ratio uses the Padé fit to ABCD purity as the denominator.

| $p_T$ | Truth Purity | | ABCD Purity | | Ratio | |
|---|---|---|---|---|---|---|
| (GeV) | Old | New | Old | New | Old | New |
| 9 | 0.566 | 0.576 | 0.370 | 0.351 | 1.505 | **1.702** |
| 11 | 0.576 | 0.609 | 0.460 | 0.453 | 1.206 | **1.258** |
| 13 | 0.620 | 0.632 | 0.654 | 0.603 | 1.112 | **1.080** |
| 15 | 0.594 | 0.637 | 0.640 | 0.659 | 0.952 | **0.966** |
| 17 | 0.584 | 0.647 | 0.630 | 0.714 | 0.862 | **0.904** |
| 19 | 0.622 | 0.714 | 0.620 | 0.741 | 0.860 | **0.938** |
| 21 | 0.667 | 0.739 | 0.700 | 0.820 | 0.875 | **0.927** |
| 23 | 0.679 | 0.713 | 0.804 | 0.857 | 0.852 | **0.862** |
| 25 | 0.651 | 0.732 | 0.560 | 0.753 | 0.787 | **0.859** |
| 27 | 0.707 | 0.756 | 0.915 | 0.869 | 0.829 | **0.865** |
| 30 | 0.679 | 0.761 | 0.712 | 0.894 | 0.766 | **0.845** |
| 34 | 0.849 | 0.777 | 0.889 | 0.741 | 0.919 | **0.837** |

The sign flip persists: the correction crosses 1.0 at $p_T \approx 14$ GeV in both configurations. At low $p_T$, the correction is *larger* (1.70 vs. 1.51 at 9 GeV). At high $p_T$, the correction is more stable (0.84–0.97 vs. 0.77–0.92), with reduced bin-to-bin fluctuation.

### 7.2 Updated $R_{\text{true}}$

Table 6: $R_{\text{true}}$ comparison: old vs. new non-tight BDT bound.

| $p_T$ bin (GeV) | $R_{\text{true}}$ (Old) | $R_{\text{true}}$ (New) | $\Delta R$ |
|---|---|---|---|
| $[8, 10]$ | 0.70 | $0.675 \pm 0.15$ | $-0.03$ |
| $[10, 12]$ | 0.80 | $0.729 \pm 0.08$ | $-0.07$ |
| $[12, 14]$ | 1.07 | $0.904 \pm 0.15$ | $-0.17$ |
| $[14, 16]$ | 1.14 | $1.058 \pm 0.21$ | $-0.08$ |
| $[16, 18]$ | 1.16 | $1.261 \pm 0.35$ | $+0.10$ |
| $[18, 20]$ | 0.98 | $0.991 \pm 0.48$ | $+0.01$ |
| $[20, 22]$ | 1.04 | $1.273 \pm 0.70$ | $+0.23$ |
| $[22, 24]$ | 1.59 | $1.779 \pm 0.41$ | $+0.19$ |
| $[24, 26]$ | 0.76 | $1.152 \pm 0.39$ | $+0.39$ |
| $[26, 28]$ | 2.09 | $1.617 \pm 0.86$ | $-0.47$ |
| $[28, 32]$ | 0.93 | $1.854 \pm 0.99$ | $+0.92$ |
| $[32, 36]$ | 1.07 | $0.827 \pm 0.43$ | $-0.24$ |

The crossing point shifted from $[12, 14]$ to $[14, 16]$ GeV. Low-$p_T$ bins are further below 1.0; high-$p_T$ bins exhibit larger positive deviations with significant statistical uncertainty.

### 7.3 Isolation Fraction Ratio: ABCD Independence Test

The key diagnostic for ABCD independence is the isolation fraction ratio for background jets in tight vs. non-tight BDT regions:

$$\rho = \frac{f_{\text{iso}}^{\text{tight}}}{f_{\text{iso}}^{\text{non-tight}}}, \qquad f_{\text{iso}} = \frac{N_{\text{iso}}}{N_{\text{iso}} + N_{\text{non-iso}}}. \tag{2}$$

ABCD independence requires $\rho = 1$. Table 7 shows the results from jet-only MC (pure background, no signal photons).

Table 7: Isolation fraction ratio $\rho$ for background jets: old (fixed non-tight = 0.50) vs. new ($E_T$-dependent non-tight). Values from jet-only MC (`MC_efficiency_jet_bdt_nom.root`).

| $p_T$ bin | BDT bound | $f_{\text{iso}}^{\text{tight}}$ | $f_{\text{iso}}^{\text{NT}}$ | $\rho$ (New) | $\rho$ (Old) |
|---|---|---|---|---|---|
| $[8, 10]$ | 0.665 | 0.526 | 0.431 | **1.22** | $\sim$0.77 |
| $[10, 12]$ | 0.635 | 0.539 | 0.403 | **1.34** | $\sim$0.85 |
| $[12, 14]$ | 0.605 | 0.560 | 0.357 | **1.57** | $\sim$1.00 |
| $[14, 16]$ | 0.575 | 0.542 | 0.311 | **1.74** | $\sim$1.10 |
| $[16, 18]$ | 0.545 | 0.539 | 0.281 | **1.92** | $\sim$1.20 |
| $[18, 20]$ | 0.515 | 0.533 | 0.264 | **2.02** | $\sim$1.30 |
| $[20, 22]$ | 0.485 | 0.545 | 0.222 | **2.46** | $\sim$1.40 |
| $[22, 24]$ | 0.455 | 0.580 | 0.215 | **2.69** | $\sim$1.59 |
| $[24, 26]$ | 0.425 | 0.560 | 0.283 | **1.98** | $\sim$0.76 |
| $[26, 28]$ | 0.395 | 0.603 | 0.240 | **2.52** | $\sim$1.77 |
| $[28, 32]$ | 0.350 | 0.625 | 0.213 | **2.94** | $\sim$0.93 |
| $[32, 36]$ | 0.290 | 0.630 | 0.355 | **1.77** | $\sim$1.07 |

**Summary:** The old configuration had $\rho \in [0.76,\ 1.77]$ (straddling 1.0); the new configuration

gives $\rho \in [1.22, 2.94]$ (entirely above 1.0, wider spread). The ABCD independence violation is *worse*, not better.

**Note:** The "Old" $\rho$ values in Table 7 are approximate reconstructions from the original investigation; the old ROOT result files have been overwritten by the current configuration. Both old and new results use the same underlying MC event samples—only the non-tight BDT cut definition changed.

### 7.4 Background Isolation $E_T$ Distributions

Figure 3 shows the normalized isolation $E_T$ distributions for *true background* clusters (truth-matched non-signal) in the tight and non-tight BDT regions. The background-only distributions are obtained by projecting `h_tight_recoisoET_background_0` and `h_nontight_recoisoET_background_0` (TH2D, filled only for clusters whose truth-matched particle is not an isolated prompt photon). In all 12 $p_T$ bins, the shapes differ, confirming that photon-like background jets are intrinsically more isolated than non-photon-like jets — a genuine BDT–isolation correlation in the background.

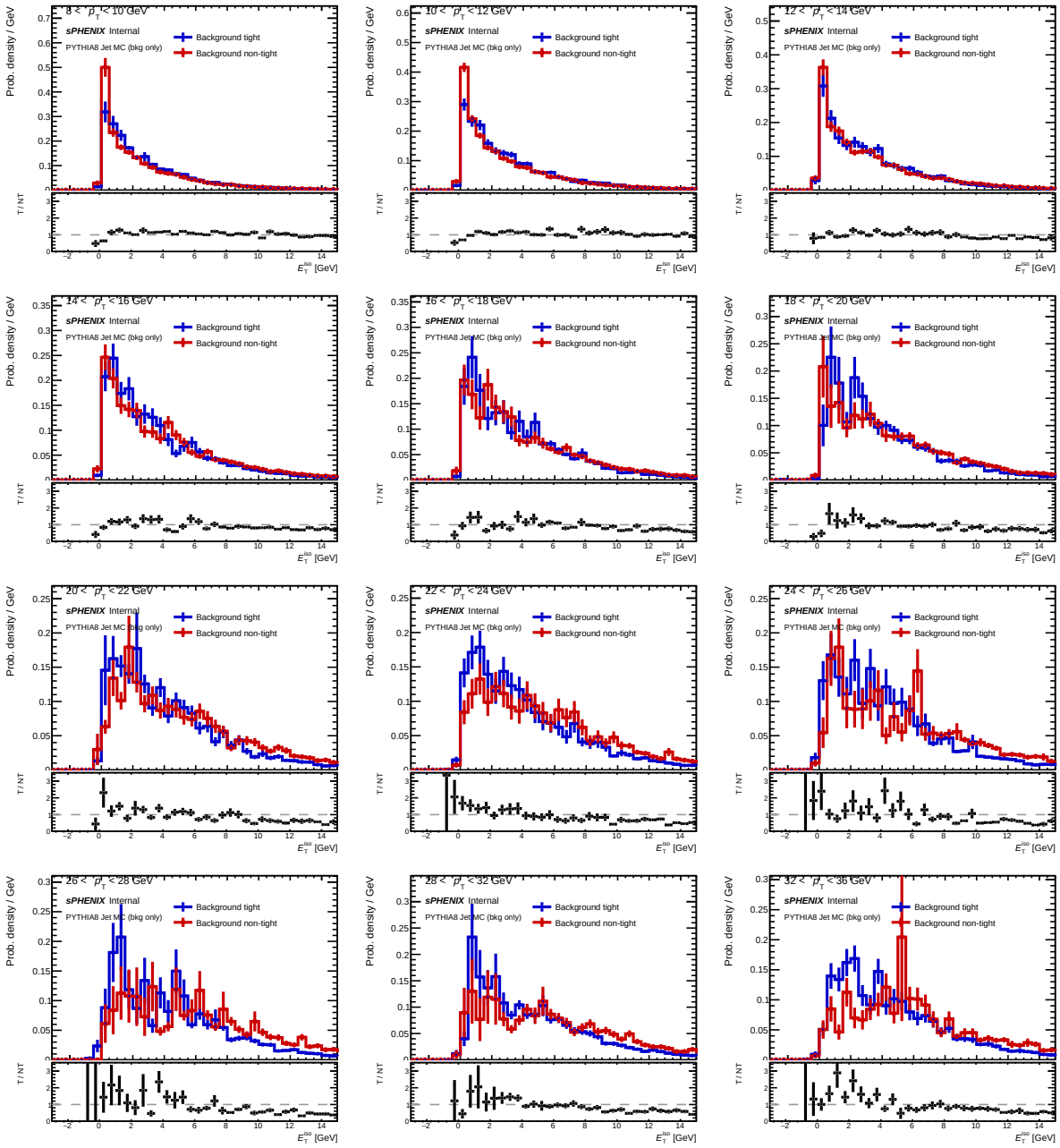


Figure 3: Normalized isolation $E_T$ distribution for **true background** (truth-matched non-signal) clusters in tight (blue) and non-tight (red) BDT regions, per $p_T$ bin. Source: inclusive MC (`MC_efficiency_bdt_nom.root`), background-only TH2D projections.

## 7.5 Physical Interpretation

The dead zone hypothesis (Section 3.2) predicted that eliminating the gap between tight and non-tight BDT regions would improve ABCD independence. Instead, the violation worsened. This demonstrates that the **primary mechanism** is intrinsic jet physics: jets that fake isolated photons (narrow, single-fragment) deposit most energy in one cluster, leaving less ambient energy in the isolation cone. The BDT selects for exactly this topology, creating a built-in BDT–isolation correlation independent of the cut geometry.

The old dead zone partially masked this correlation at low $p_T$ by excluding intermediate-BDT jets from the non-tight region. These jets had isolation fractions between the tight and non-tight extremes. Their exclusion pushed $f_{\text{iso}}^{\text{NT}}$ higher at low $p_T$, making $\rho$ closer to 1.0. Closing the gap

pulls them back in, revealing the full correlation.

### 7.6 Updated Signal Leakage Fractions

Table 8: Signal leakage fractions: old vs. new non-tight BDT bound.

| $p_T$ bin | $c_B$ (Old) | $c_B$ (New) | $c_C$ (Old) | $c_C$ (New) | $c_D$ (Old) | $c_D$ (New) |
|---|---|---|---|---|---|---|
| $[8, 10]$ | 5–6% | 5.7% | 2.6–3% | 6.7% | <0.6% | 0.7% |
| $[12, 14]$ | ∼7% | 6.6% | ∼3.5% | 5.7% | <0.6% | 0.8% |
| $[16, 18]$ | ∼8% | 7.2% | ∼4% | 4.9% | <0.6% | 0.6% |
| $[20, 22]$ | ∼9% | 7.5% | ∼4% | 4.4% | <0.6% | 0.5% |
| $[26, 28]$ | ∼11% | 8.8% | ∼4.5% | 4.3% | <0.6% | 0.2% |
| $[32, 36]$ | ∼13% | 8.4% | ∼4.6% | 2.2% | <0.6% | 0.2% |

$c_C$ increased at low $p_T$ (from ∼3% to 6.7% at 8–10 GeV) because the non-tight region now extends to higher BDT scores (up to 0.68 vs. 0.50 at $E_T = 8$ GeV), capturing more signal photons just below the tight threshold. $c_B$ decreased at high $p_T$ (from ∼13% to 8.4% at 32–36 GeV), because signal photons formerly lost in the dead zone now correctly enter the tight region. $c_D$ is essentially unchanged (below 1% in both configurations).

### 7.7 Signal Contamination Correction

*Added April 8, 2026.* A subsequent investigation revealed that the jet MC samples (jet10, jet15, jet20, jet30, jet50) contain a large fraction of truth-matched prompt photons that pass all analysis cuts. These signal photons dominate region A (tight, isolated), comprising 58–78% of the jet MC total there. Table 9 summarizes the signal fraction in region A per $p_T$ bin.

Table 9: Signal fraction (truth-matched prompt photons) in the jet MC region A (tight, isolated) per $p_T$ bin. Signal fractions in other regions are lower: B (4–11%), C (2–21%), D ($< 1$%).

| $p_T$ bin (GeV) | Signal fraction in A |
|---|---|
| $[8, 10]$ | 57.6% |
| $[10, 12]$ | 60.9% |
| $[12, 14]$ | 63.2% |
| $[14, 16]$ | 63.7% |
| $[16, 18]$ | 64.7% |
| $[18, 20]$ | 71.4% |
| $[20, 22]$ | 73.9% |
| $[22, 24]$ | 71.3% |
| $[24, 26]$ | 73.2% |
| $[26, 28]$ | 75.6% |
| $[28, 32]$ | 76.1% |
| $[32, 36]$ | 77.7% |

This contamination has important consequences for the $R$-factor interpretation. Two distinct $R$ factors can be computed from jet MC:

- $R_{\text{raw}}$ (signal-contaminated): uses all jet MC counts without truth subtraction. Always $> 1$ (range 1.47–6.16), monotonically increasing.

- $R_{\text{bkg}}$ (= `h_R`, pure background): after subtracting truth-matched signal from each region. Crosses 1.0 at $p_T \approx 14$ GeV (range 0.675–1.854).

Table 10 compares the two.

Table 10: $R$-factor decomposition: signal-contaminated ($R_{\text{raw}}$, from jet MC totals) vs. pure background ($R_{\text{bkg}} =$ `h_R`, after truth subtraction). The earlier analysis incorrectly used $R_{\text{raw}}$ as "pure background $R$."

| $p_T$ bin (GeV) | $R_{\text{raw}}$ (contaminated) | $R_{\text{bkg}}$ (`h_R`, pure bkg) |
|---|---|---|
| $[8, 10]$ | 1.47 | 0.675 |
| $[10, 12]$ | 1.73 | 0.729 |
| $[12, 14]$ | 2.29 | 0.904 |
| $[14, 16]$ | 2.62 | 1.058 |
| $[16, 18]$ | 3.00 | 1.261 |
| $[18, 20]$ | 3.19 | 0.991 |
| $[20, 22]$ | 4.21 | 1.273 |
| $[22, 24]$ | 5.03 | 1.779 |
| $[24, 26]$ | 3.22 | 1.151 |
| $[26, 28]$ | 4.82 | 1.617 |
| $[28, 32]$ | 6.16 | 1.854 |
| $[32, 36]$ | 3.08 | 0.827 |

The factor-of-2-to-6 difference between $R_{\text{raw}}$ and $R_{\text{bkg}}$ is entirely due to signal contamination inflating region A. The earlier isoET overlay plots (Figure 3) used the total jet MC in the tight region, where the sharp isolation peak near 1 GeV was dominated by signal photons, not background. After truth subtraction, the background-only isoET distributions (Figure 5) show that tight and non-tight background have nearly identical isolation at low $p_T$ (8–14 GeV), with only a subtle divergence at high $p_T$.

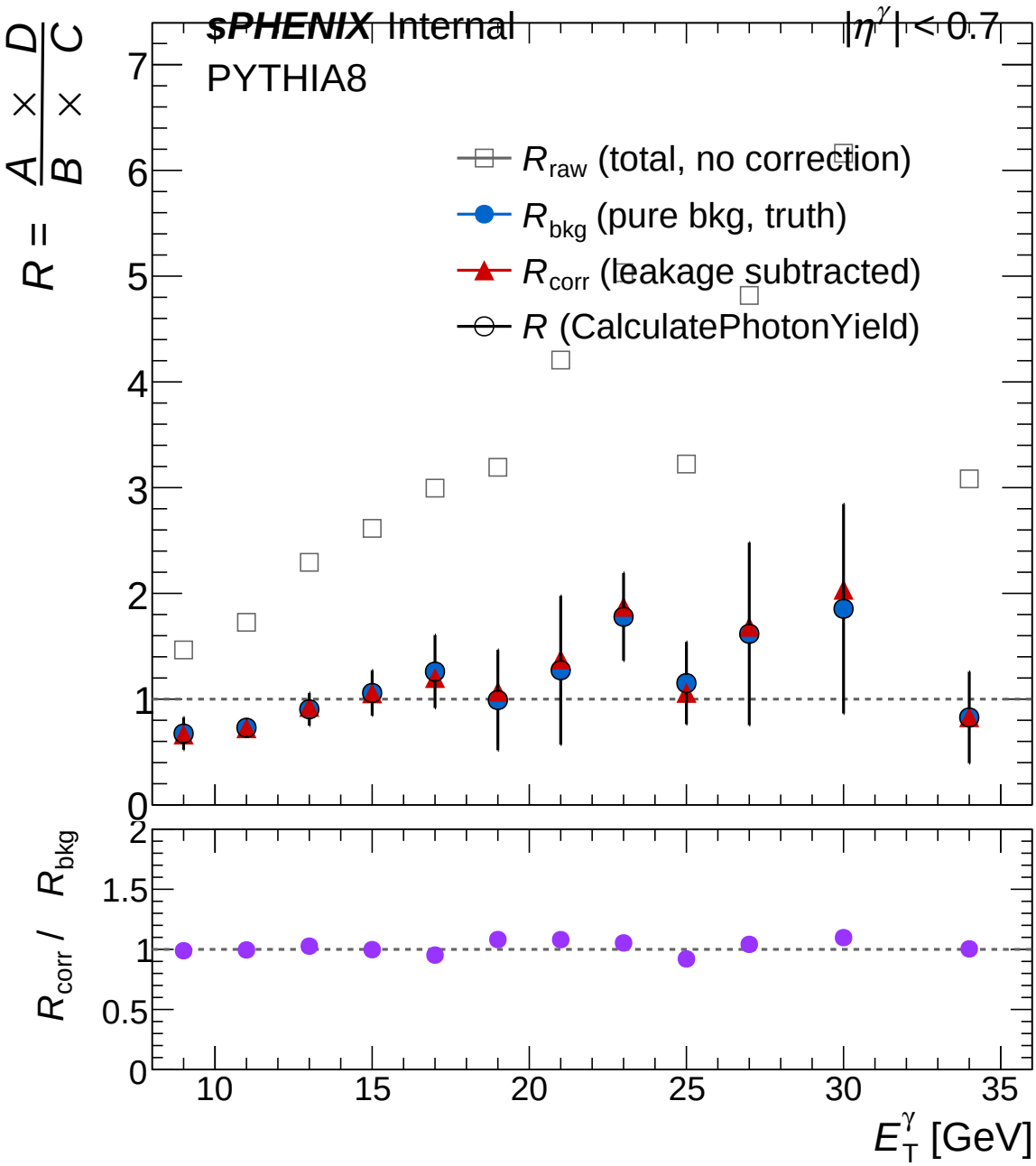


Figure 4: $R$-factor decomposition vs. $p_T$: signal-contaminated $R_{\text{raw}}$ (from jet MC totals), pure background $R_{\text{bkg}}$ (= `h_R`), and jet MC signal fraction in region A. Produced with `plot_R_decomposition.C`.

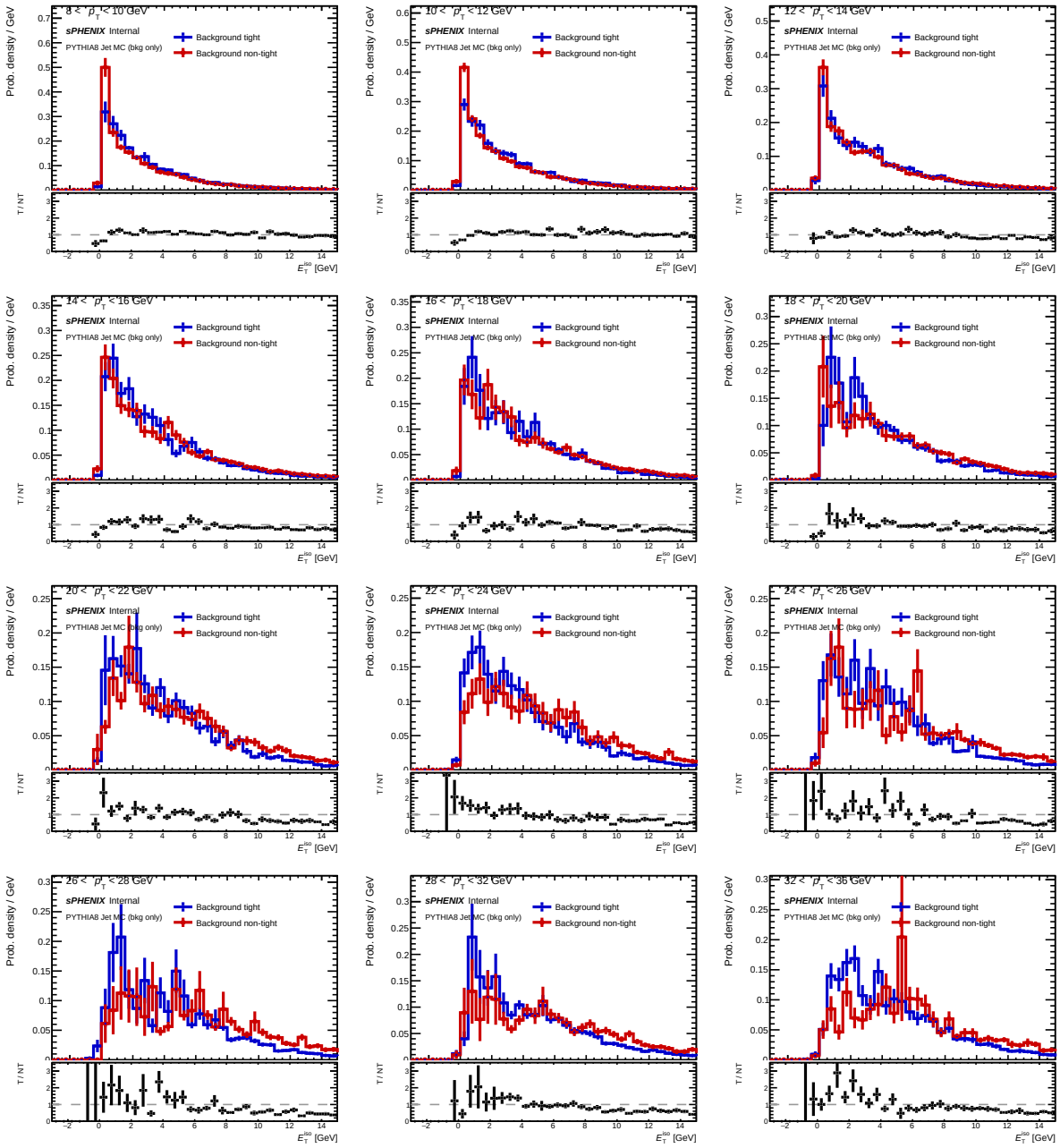


Figure 5: Normalized isolation $E_T$ distribution for **pure background** (after truth signal subtraction) in tight and non-tight BDT regions, per $p_T$ bin. Compare with Figure 3, which used signal-contaminated jet MC totals. Produced with `plot_bkg_isoET_tight_vs_nontight.C`.

**Corrected interpretation:** The $R_{\text{bkg}}$ crossover at $p_T \approx 14$ GeV is a genuine but subtle background physics effect. At low $p_T$, tight and non-tight background jets have nearly identical isolation properties ($R \approx 1$, slightly below). At high $p_T$, photon-like background jets become marginally more isolated, pushing $R$ above 1. The earlier conclusion that the "pure background" $R$ is always $> 1$ (1.4–5.0 range) and that the sign flip in `h_R` was caused by signal leakage corrections was incorrect—those values were $R_{\text{raw}}$ computed from signal-contaminated jet MC totals.

## 7.8 Updated Recommendations

In addition to the original recommendations (Section 6):

6. The purity correction sign flip is a **genuine background physics effect** ($R_{\text{bkg}}$ crossing 1.0) that cannot be eliminated by adjusting the non-tight BDT boundary. The magnitude is modest after removing signal contamination from the diagnostics.

7. **Jet MC diagnostics must account for signal contamination**: any isolation-based ABCD diagnostic computed from raw jet MC totals is dominated by the 58–78% signal fraction in region A. Always use truth-subtracted quantities (`h_R`, background-only isoET) for physics interpretation of background behavior.

8. Approximately 30 systematic variation configs still use the old fixed non-tight bound (`bdt_max` = 0.50, missing `bdt_max_slope`/`bdt_max_intercept` keys). Regenerate if the $E_T$-dependent bound is intended for all variations.

9. `RecoEffCalculator.C` (old code path) does not read the parametric non-tight fields—only `RecoEffCalculator_TTreeReader.C` does. Confirm which code path is active in production.

# A Plots

The following diagnostic plots were produced by the investigation agents.

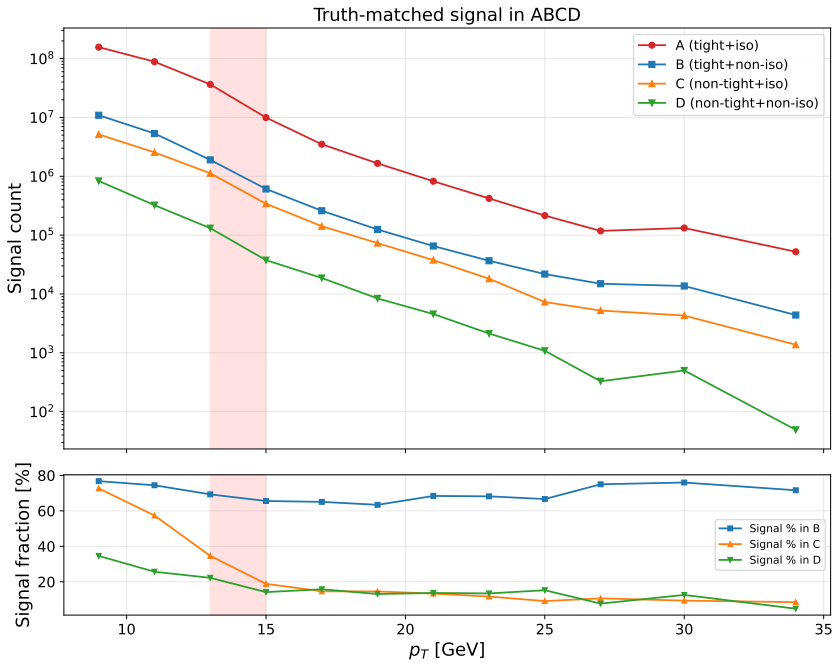


Figure 6: Signal counts in each ABCD region vs. $p_T$.

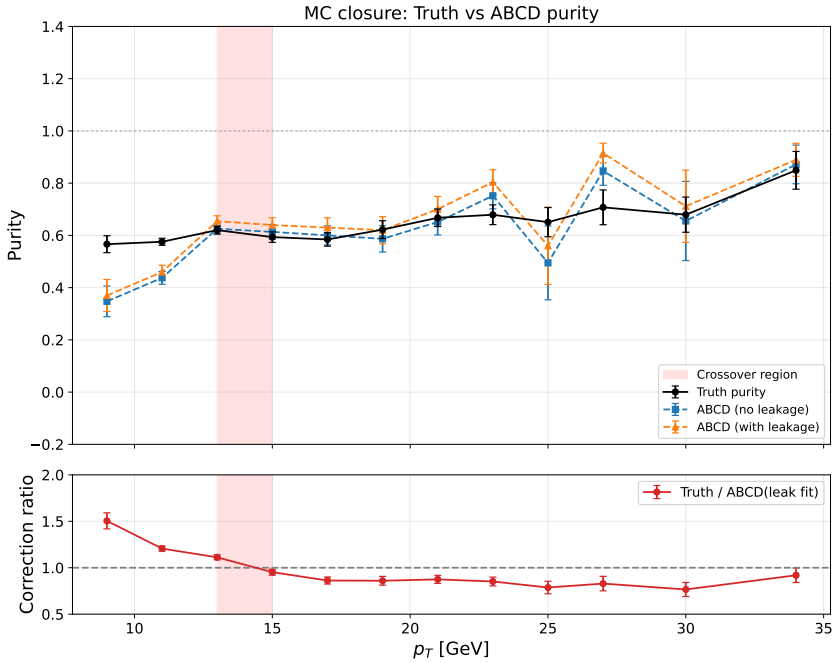


Figure 7: Truth purity vs. leakage-corrected ABCD purity as a function of $p_T$.

Additional multi-page diagnostics are in `plotting/figures/purity_fit_diagnostics.pdf` (Padé fit curves, residuals, ratio crossing analysis).

**Note (April 8, 2026):** The diagnostic plots in this appendix (Figures 6–7) were generated with the *old* fixed non-tight BDT configuration. The underlying ROOT result files have since been overwritten with the current $E_T$-dependent configuration. To regenerate the old plots, one would need to rerun the pipeline with `bdt_max_slope: 0` and `bdt_max_intercept: 0.5` in the non-tight block.